

1. Answer BOTH parts (a) and (b).

- (a) Discuss the meaning of the terms *shielding*, *penetration* and *effective nuclear charge* in the context of polyelectronic atoms. [4]
- (b) Sketch radial distribution functions for the 1s, 2s and 2p orbitals, and use them to rationalise the ground-state electronic configuration of Li. [6]